

v4 defringe · one mask per frame

Agreed design — not coded yet. Does not overwrite defringe_v22 or v3. notes/V4_PIPELINE.md

1. Raw frame · hints only (shutter span, catalog/PMT, last-frame mask). Never a locked stack q.

2a. FFT lines

- fy ridges + fx columns (fy=0 OK)
- harmonics as extras
- leftover peaks after a notch
- chirp = nearby edge bins, not family 2

2b. Linescans

- H, V, both diagonals always
- One (qy, qx) or none
- none is honest (shutter 2-D ridge)
- $P(x) / L-C-R$ = chirp extras

3. Rank into one list · core (both views agree) → agreed extras (harmonics, chirp edges) → dubious (one view / leftover)
If congruence is none: core = loudest symmetric FFT ridge. Do not blend fy and fx into one gate.

4. Soft recursion on ONE mask (support + strength)

Start: core only, low α . Each step: add the next line and/or raise α a little. Notch.
Look at predicted (IFFT of attenuated bins), removed (raw—cleaned), leftover, and the increment.
Live: if removed looks like cells OR removed diverges from predicted → undo this step only.
Shutter: no biology brake; push until leftover is not fringe. Predicted must still look like fringe.

5. Write frame · cleaned + removed + predicted + applied heatmap + rung log
This mask may hint the next frame. It does not walk q for the rest of the stack.
Outputs: <channel>/defringe_v4/ stacks, per_frame.csv, overview.pdf

What 'better' looks like at each rung

Predicted fringe = IFFT of energy the current mask attenuates.

As lines are added it should look more complete, still striped, still conjugate-symmetric.

Removed should track that image. Leftover should lose fringe, not cells.

Internal checks (every step, not only the PDF):

- predicted looks like fringe (not soma, not a full-FOV bandpass)
- removed $\approx$ predicted (biology sneaking in shows up as extra blobs in removed)
- increment = removed_k – removed_{k-1} is the thing we undo if it fails

Do not IFFT whole FFT rows/columns — that bandpasses the FOV and looks like 'predicted cells'.

Steal pack_D's thin ridges and local excess attenuation; do not steal union-of-families apply.

α rungs: few (≈ 3 live, ≈ 4 shutter). Recursion budget goes to adding lines in certainty order.

overview.pdf · how we evaluate a run

1. Cover — shutter span, empty vs core-only vs full masks, RMS, n biology-brakes undone
2. Shutter detect — FOV std cliff (same method as v3)
3. Stack traces — lines in mask, max α , removed RMS, predicted↔removed agreement, brake
4. Means — raw / cleaned / removed / predicted
5. Rung story — shutter mid, anchors 160 / 700 / 1061, strong / empty / brake / chirp frames:
ranked lines · mask after each kept step · predicted | removed | leftover | cleaned
plus any undone increment (pushed too hard)
6. Undone / discarded gallery
7. Linescan vs FFT on those frames (four traces + congruent peaks vs FFT core)

Judge: predicted more complete and still a fringe; removed follows it;
if removed grows cells that predicted does not have, that step failed.

Do not carry these forward from v3

- Stack q-tracker (± 10 / 50 frames, apply ± 2) dominating linescan/FFT
- Union-apply of frozen pack_D families as the product
- Recursion = next unused ($\bar{a}$ axis, q) instead of growing one mask
- Starting at full α so the first notch can eat cells and 'cancel' the clean
- Independent tile cleans · IFFT of whole rows/columns
- Treating edge-q as a second family · blending fy/fx into one gate
- Flattening shutter 756-760 as the definition of success
- Overwriting defringe_v22 or promoting Haj Grant ChanA until 160 cleans from this frame's core and 700 is either a real family or judged empty